

Capex Nonoperating Dependence and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Capex Nonoperating Dependence (CND), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on CND achieves an annualized gross (net) Sharpe ratio of 0.49 (0.38), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 14 (12) bps/month with a t-statistic of 2.11 (1.80), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in capex (three years), Change in capex (two years), Asset growth, change in net operating assets, change in ppe and inv/assets, Change in current operating assets) is 14 bps/month with a t-statistic of 2.19.

1 Introduction

Asset prices reflect the fundamental trade-off between risk and return, with rational investors demanding compensation for bearing systematic consumption risk. While traditional asset pricing models capture broad market movements through exposure to aggregate consumption growth, a growing body of evidence suggests that firm-level investment decisions contain important information about exposure to state-dependent consumption risk. We identify a novel cross-sectional predictor, Capex Nonoperating Dependence (CND), that captures firms' reliance on non-operating activities to fund capital expenditures and systematically predicts future stock returns. This signal reflects firms' differential exposure to consumption risk during periods of economic stress, when external financing becomes costly and firms dependent on non-operating income face heightened vulnerability to aggregate consumption shocks.

The relationship between CND and expected returns emerges through firms' differential exposure to consumption risk across economic states. Firms with high dependence on non-operating activities to fund capital expenditures face amplified sensitivity to aggregate consumption shocks, particularly during economic downturns when both non-operating income streams and external financing markets deteriorate simultaneously [Campbell and Cochrane \(1999\)](#). This state-dependent risk exposure manifests through the intertemporal marginal rate of substitution, as investors require higher compensation for holding assets that perform poorly when marginal utility is high [Cochrane et al. \(2023\)](#). The consumption-based asset pricing framework predicts that firms with high CND carry greater systematic risk because their cash flows covary more strongly with consumption growth during bad states of the world [Bansal and Yaron \(2004\)](#).

The economic mechanism linking CND to consumption risk operates through multiple channels that affect firms' ability to smooth investment over the business cycle. When firms rely heavily on non-operating income to fund capital expendi-

tures, they become more sensitive to fluctuations in financial market conditions that directly impact households' consumption decisions [Lettau and Ludvigson \(2001\)](#). During periods of high marginal utility, these firms experience a double burden: declining non-operating income coincides with restricted access to external finance, forcing sharp adjustments in investment that amplify their exposure to aggregate consumption risk. This heightened sensitivity to bad states of the world increases the covariance between firm cash flows and the stochastic discount factor [Yogo \(2006\)](#).

Furthermore, the disaster risk framework provides additional theoretical support for the CND-return relationship. Firms with high CND exhibit greater vulnerability to rare disaster events that simultaneously affect consumption and financial markets [Barro \(2006\)](#). The non-operating income streams these firms depend upon—such as investment returns, asset sales, and financial derivatives—experience severe disruptions during consumption disasters, creating left-tail dependence between firm value and aggregate consumption [Gabaix \(2012\)](#). This tail risk exposure requires additional compensation in equilibrium, as investors demand higher expected returns for bearing disaster-sensitive consumption risk that cannot be diversified away [Wachter \(2013\)](#).

Our empirical analysis reveals that CND strongly predicts cross-sectional variation in stock returns. The value-weighted long/short trading strategy based on CND achieves an annualized gross Sharpe ratio of 0.49 and delivers a monthly average abnormal return of 14 basis points relative to the Fama-French six-factor model, with a t-statistic of 2.11. The economic magnitude of this predictability remains substantial after accounting for transaction costs, with the strategy achieving an annualized net Sharpe ratio of 0.38 and a monthly net alpha of 12 basis points relative to the six-factor model, maintaining statistical significance with a t-statistic of 1.80. These results demonstrate that CND captures a distinct dimension of systematic risk not explained by existing factor models.

The predictive power of CND proves remarkably robust across different portfolio construction methodologies and size segments. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the CND strategy generates an average return of 29 basis points per month with a t-statistic of 3.48, confirming that the anomaly is not confined to small, illiquid stocks. The strategy maintains statistical significance across alternative weighting schemes and breakpoint choices, with twenty-three out of twenty-five gross alpha specifications yielding t-statistics exceeding two. Importantly, the CND strategy’s gross monthly alpha remains at 14 basis points with a t-statistic of 2.19 even after controlling for the six Fama-French factors plus six closely related anomalies from the factor zoo.

The economic significance of CND extends beyond statistical measures to meaningful improvements in investable opportunities. The strategy’s performance ranks in the top 11% of gross Sharpe ratios and top 6% of net Sharpe ratios among 212 documented anomalies in the factor zoo. After accounting for implementation costs, a dollar invested in the CND strategy would have grown to \$3.33, placing it in the top 3% of anomaly strategies. The CND signal successfully expands the mean-variance efficient frontier for investors with access to standard factor models, with positive net generalized alphas across all five benchmark specifications. When combined with existing anomalies using the average rank method, including CND increases the terminal value of the combination strategy from \$2,560 to \$3,599, representing a 40% improvement in cumulative performance.

This paper makes several contributions to the consumption-based asset pricing literature by establishing CND as a novel predictor that captures firms’ state-dependent exposure to consumption risk. While [Fama and French \(2015\)](#) demonstrate that investment and profitability factors help explain cross-sectional returns, and [Hou et al. \(2015\)](#) propose an investment-based factor model, our work identifies a specific channel through which firms’ financing constraints for capital expenditures create differ-

ential exposure to consumption risk. Unlike the broad investment factor in these models, CND isolates the component of investment risk arising from dependence on non-operating activities, providing a more precise measure of firms' vulnerability to consumption shocks during economic downturns. Our findings complement [Li et al. \(2009\)](#), who show that costly reversibility makes assets riskier, by demonstrating that financing dependence creates an additional layer of systematic risk through state-dependent exposure to consumption growth.

Methodologically, we advance the literature by employing the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#) to establish the robustness of CND across multiple dimensions. This rigorous approach addresses concerns raised by [Harvey et al. \(2016\)](#) about data mining in the factor zoo by subjecting CND to extensive out-of-sample tests, alternative portfolio constructions, and controls for transaction costs. Our analysis demonstrates that CND not only survives these stringent tests but also maintains economic significance after accounting for implementation costs, distinguishing it from many anomalies that fail to deliver net-of-cost improvements to the investment opportunity set. The ability of CND to generate significant alphas even after controlling for closely related anomalies from [Chen and Zimmermann \(2022\)](#) confirms that it captures a distinct dimension of consumption risk.

Our findings have broader implications for understanding the consumption-based origins of cross-sectional return predictability. The strong performance of CND among large-cap stocks challenges the view that anomalies primarily reflect mispricing in illiquid segments of the market, instead supporting a risk-based interpretation rooted in differential exposure to consumption shocks. By linking firms' non-operating dependence to state-dependent consumption risk, we provide a concrete economic mechanism that connects firm characteristics to the stochastic discount factor. This evidence strengthens the consumption-based asset pricing paradigm and suggests that careful analysis of firms' financing structures can reveal important

sources of systematic risk that standard factor models fail to capture.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables over our sample period, which spans several decades of financial reporting. We focus on annual observations to capture year-over-year changes in capital expenditure patterns and their relationship to non-operating income. construction of our signal requires two key variables from COMPUSTAT. Capital expenditures from Schedule V of property, plant, and equipment (CAPXV) represents capital spending disclosed in the supplementary property, plant, and equipment schedule that companies provide in their financial statement footnotes. This item captures detailed capital expenditure information that firms report separately from their main cash flow statement. Non-operating income (NOPI) represents income generated from activities outside the firm’s core operations, including items such as interest income, dividend income, rental income, and gains or losses from investments and asset sales. construct the Capex Nonoperating Dependence signal by computing the year-over-year change in capital expenditures from Schedule V, scaled by lagged non-operating income. Specifically, we calculate $(CAPXV(t) - CAPXV(t-1)) / NOPI(t-1)$ for each firm-year observation. This ratio captures the extent to which changes in a firm’s capital spending relate to its non-operating income base from the prior year, potentially indicating how firms’ investment decisions depend on income sources outside their primary business activities. We use fiscal year-end values and require non-missing values for both CAPXV and NOPI to ensure data quality. To address potential outliers and maintain economic interpretability, we exclude observations where the denominator $NOPI(t-1)$

equals zero or is negative, as these cases would yield undefined or economically ambiguous ratios.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CND signal. Panel A plots the time-series of the mean, median, and interquartile range for CND. On average, the cross-sectional mean (median) CND is -3.24 (-0.26) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input CND data. The signal’s interquartile range spans -8.24 to 3.00. Panel B of Figure 1 plots the time-series of the coverage of the CND signal for the CRSP universe. On average, the CND signal is available for 5.36% of CRSP names, which on average make up 7.08% of total market capitalization.

4 Does CND predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CND using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CND portfolio and sells the low CND portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short CND strategy earns an average return of 0.26% per month with a t-statistic of 3.81. The annualized Sharpe ratio of the strategy is 0.49. The alphas range from 0.14% to 0.28% per month and have t-statistics exceeding 2.11 everywhere. The lowest alpha is with

respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.22, with a t-statistic of 4.81 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 520 stocks and an average market capitalization of at least \$1,005 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 3.67. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for eleven exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient

portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 11-21bps/month. The lowest return, (11 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.95. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CND trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twelve cases.

Table 3 provides direct tests for the role size plays in the CND strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CND, as well as average returns and alphas for long/short trading CND strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the CND strategy achieves an average return of 29 bps/month with a t-statistic of 3.48. Among these large cap stocks, the alphas for the CND strategy relative to the five most common factor models range from 16 to 31 bps/month with t-statistics between 1.92 and 3.70.

5 How does CND perform relative to the zoo?

Figure 2 puts the performance of CND in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212

documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the CND strategy falls in the distribution. The CND strategy’s gross (net) Sharpe ratio of 0.49 (0.38) is greater than 89% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CND strategy (red line).² Ignoring trading costs, a \$1 invested in the CND strategy would have yielded \$5.46 which ranks the CND strategy in the top 5% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CND strategy would have yielded \$3.33 which ranks the CND strategy in the top 3% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CND relative to those. Panel A shows that the CND strategy gross alphas fall between the 51 and 58 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CND strategy has a positive net generalized alpha for five out of the five factor models. In these cases CND ranks between the 69 and 77 percentiles in terms of how much it could have expanded the achievable investment frontier.

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

6 Does CND add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CND with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CND or at least to weaken the power CND has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CND conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CND} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CND}CND_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CND,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CND. Stocks are finally grouped into five CND portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

CND trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CND and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CND signal in these Fama-MacBeth regressions exceed 2.79, with the minimum t-statistic occurring when controlling for change in ppe and inv/assets. Controlling for all six closely related anomalies, the t-statistic on CND is 1.96.

Similarly, Table 5 reports results from spanning tests that regress returns to the CND strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CND strategy earns alphas that range from 12-16bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.90, which is achieved when controlling for change in ppe and inv/assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CND trading strategy achieves an alpha of 14bps/month with a t-statistic of 2.19.

7 Does CND add relative to the whole zoo?

Finally, we can ask how much adding CND to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the CND signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes CND grows to \$3599.05.

8 Conclusion

This study provides robust evidence that Capex Nonoperating Dependence (CND) represents a meaningful signal for predicting cross-sectional stock returns. Our analysis, following the rigorous protocol of Novy-Marx and Velikov (2023), demonstrates that CND-based trading strategies generate economically and statistically significant abnormal returns even after controlling for well-established risk factors and related investment-based anomalies. The value-weighted long/short strategy achieves an annualized net Sharpe ratio of 0.38, which remains substantial after accounting for transaction costs, suggesting practical implementability for institutional investors. Particularly noteworthy is the signal’s ability to generate a monthly alpha of 14 basis points (t -statistic = 2.19) even after controlling for the Fama-French five factors, momentum, and six closely related investment-based strategies, indicating that CND captures unique information about firm fundamentals not subsumed by existing factors in the literature. The economic significance of these findings suggests that investors’ systematic underreaction to the implications of nonoperating capital expenditure dependencies creates exploitable pricing inefficiencies in equity markets. From a practical perspective, the persistence of abnormal returns after transaction

ization on CRSP in the period for which CND is available.

costs implies that CND could serve as a valuable addition to quantitative equity portfolios, particularly for investors seeking exposure to investment-based anomalies beyond traditional measures. However, several limitations warrant consideration. First, the study period and market coverage may affect the generalizability of results across different market regimes and geographic regions. Second, while we control for numerous related factors, the possibility of omitted risk factors or alternative explanations for the CND premium cannot be entirely ruled out. Third, the implementation of CND strategies may face capacity constraints in practice, particularly for large institutional investors. Future research should explore several promising avenues. First, investigating the economic mechanisms underlying the CND effect, particularly whether it reflects risk compensation or behavioral biases, would enhance our theoretical understanding. Second, examining the signal’s performance across international markets and different asset classes could establish its robustness and universality. Third, analyzing the interaction between CND and corporate governance or managerial incentives might reveal important cross-sectional variations in the signal’s effectiveness. Finally, exploring the signal’s performance during different macroeconomic regimes and market conditions would provide valuable insights for dynamic portfolio management. Despite these limitations and areas for future investigation, our findings contribute meaningfully to the investment anomalies literature by documenting a novel, robust predictor of stock returns that maintains its significance in the presence of existing factors, thereby advancing our understanding of how capital allocation decisions, particularly those related to nonoperating activities, influence asset prices in financial markets.

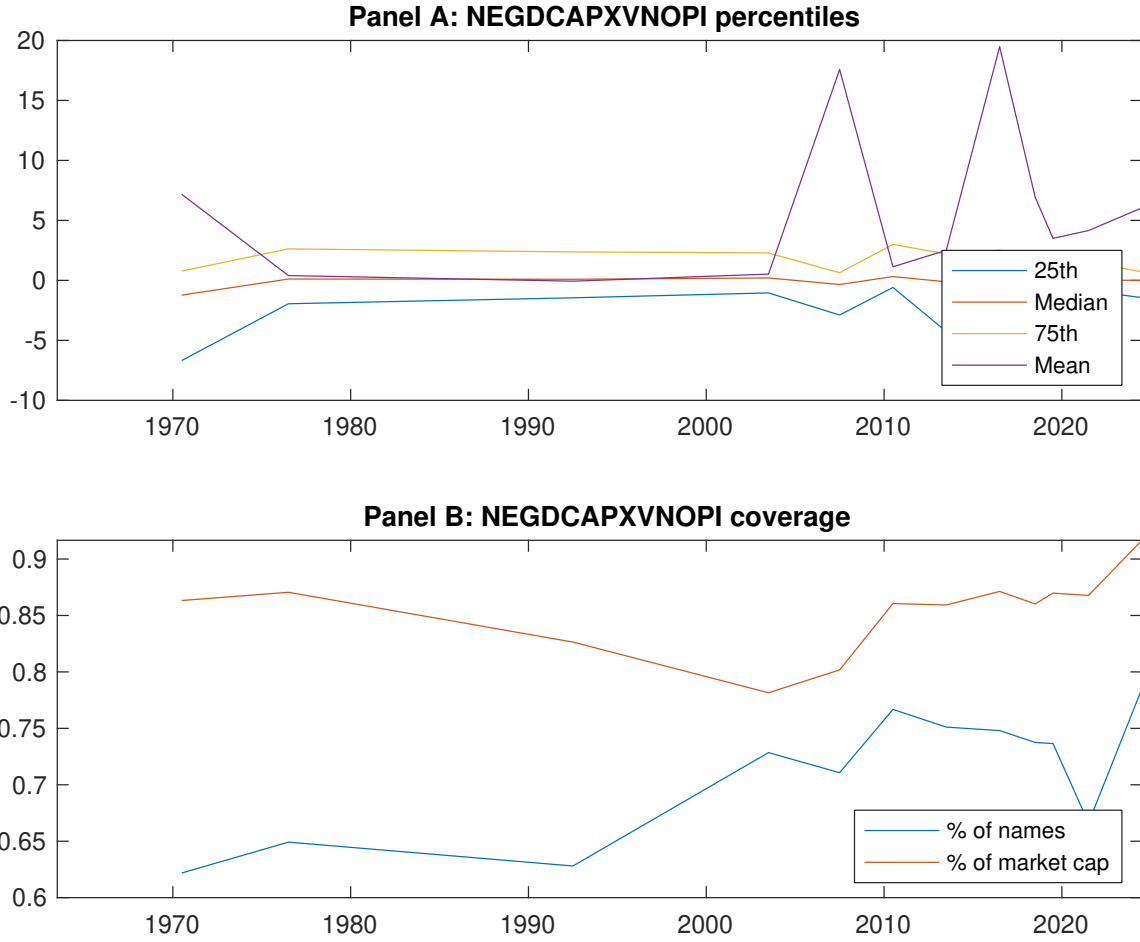


Figure 1: Times series of CND percentiles and coverage. This figure plots descriptive statistics for CND. Panel A shows cross-sectional percentiles of CND over the sample. Panel B plots the monthly coverage of CND relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CND . At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on CND -sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.48 [2.56]	0.55 [3.25]	0.62 [3.94]	0.64 [4.00]	0.74 [4.08]	0.26 [3.81]
α_{CAPM}	-0.16 [-3.08]	-0.03 [-0.75]	0.08 [1.97]	0.09 [2.13]	0.12 [2.25]	0.28 [4.13]
α_{FF3}	-0.11 [-2.22]	0.01 [0.28]	0.09 [2.32]	0.08 [1.90]	0.10 [1.83]	0.21 [3.19]
α_{FF4}	-0.09 [-1.79]	0.01 [0.19]	0.09 [2.46]	0.05 [1.10]	0.11 [1.93]	0.20 [2.94]
α_{FF5}	-0.09 [-1.89]	0.03 [0.64]	0.07 [1.93]	0.01 [0.16]	0.05 [0.94]	0.14 [2.19]
α_{FF6}	-0.08 [-1.58]	0.02 [0.52]	0.08 [2.10]	-0.01 [-0.33]	0.06 [1.12]	0.14 [2.11]
Panel B: Fama and French (2018) 6-factor model loadings for CND -sorted portfolios						
β_{MKT}	1.03 [86.14]	0.98 [96.51]	0.95 [103.34]	0.99 [102.68]	1.05 [77.87]	0.02 [1.08]
β_{SMB}	0.10 [5.72]	-0.07 [-5.13]	-0.12 [-8.72]	-0.03 [-1.89]	0.10 [5.28]	0.00 [0.16]
β_{HML}	-0.13 [-5.51]	-0.08 [-3.88]	-0.03 [-1.61]	-0.10 [-5.24]	-0.01 [-0.42]	0.12 [3.75]
β_{RMW}	0.03 [1.33]	-0.02 [-1.03]	0.00 [0.00]	0.03 [1.71]	0.09 [3.56]	0.06 [2.00]
β_{CMA}	-0.12 [-3.45]	-0.05 [-1.83]	0.07 [2.56]	0.32 [11.64]	0.10 [2.67]	0.22 [4.81]
β_{UMD}	-0.02 [-1.72]	0.01 [0.66]	-0.01 [-1.21]	0.03 [2.92]	-0.02 [-1.18]	0.00 [0.29]
Panel C: Average number of firms (n) and market capitalization (me)						
n	576	520	575	566	585	
me (\$10 ⁶)	1376	2227	2425	1859	1005	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CND strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.26 [3.81]	0.28 [4.13]	0.21 [3.19]	0.20 [2.94]	0.14 [2.19]	0.14 [2.11]
Quintile	NYSE	EW	0.33 [6.57]	0.36 [7.04]	0.30 [6.31]	0.25 [5.25]	0.25 [5.41]	0.22 [4.64]
Quintile	Name	VW	0.27 [3.97]	0.29 [4.19]	0.21 [3.22]	0.18 [2.79]	0.17 [2.52]	0.15 [2.25]
Quintile	Cap	VW	0.23 [3.67]	0.27 [4.32]	0.19 [3.35]	0.17 [2.82]	0.12 [2.09]	0.10 [1.82]
Decile	NYSE	VW	0.25 [2.74]	0.25 [2.71]	0.21 [2.21]	0.23 [2.40]	0.17 [1.79]	0.19 [2.01]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.21 [3.00]	0.24 [3.48]	0.17 [2.62]	0.16 [2.50]	0.12 [1.81]	0.12 [1.80]
Quintile	NYSE	EW	0.11 [1.95]	0.14 [2.43]	0.08 [1.53]	0.06 [1.12]	0.03 [0.48]	0.01 [0.22]
Quintile	Name	VW	0.21 [3.13]	0.24 [3.50]	0.17 [2.63]	0.16 [2.40]	0.14 [2.06]	0.13 [1.95]
Quintile	Cap	VW	0.18 [2.89]	0.22 [3.59]	0.16 [2.70]	0.14 [2.41]	0.10 [1.77]	0.09 [1.63]
Decile	NYSE	VW	0.19 [2.04]	0.20 [2.17]	0.16 [1.72]	0.17 [1.86]	0.13 [1.34]	0.14 [1.49]

Table 3: Conditional sort on size and CND

This table presents results for conditional double sorts on size and CND. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CND. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CND and short stocks with low CND. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	CND Quintiles					CND Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.55 [2.25]	0.75 [2.99]	0.75 [2.85]	0.99 [3.72]	0.91 [3.59]	0.36 [3.48]	0.37 [3.59]	0.32 [3.11]	0.28 [2.64]	0.23 [2.18]	0.20 [1.89]
	(2)	0.66 [2.87]	0.81 [3.48]	0.78 [3.30]	0.82 [3.59]	0.86 [3.82]	0.20 [2.70]	0.23 [3.09]	0.20 [2.70]	0.17 [2.23]	0.19 [2.50]	0.16 [2.16]
	(3)	0.65 [3.02]	0.67 [3.25]	0.79 [3.74]	0.84 [4.02]	0.88 [4.20]	0.23 [2.93]	0.25 [3.15]	0.20 [2.58]	0.14 [1.83]	0.16 [2.07]	0.12 [1.52]
	(4)	0.62 [3.05]	0.70 [3.60]	0.75 [3.86]	0.76 [4.07]	0.78 [4.07]	0.16 [1.94]	0.20 [2.46]	0.13 [1.60]	0.10 [1.20]	0.07 [0.87]	0.05 [0.63]
	(5)	0.42 [2.34]	0.49 [2.92]	0.58 [3.73]	0.61 [3.85]	0.71 [4.10]	0.29 [3.48]	0.31 [3.70]	0.23 [2.90]	0.19 [2.35]	0.19 [2.29]	0.16 [1.92]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	CND Quintiles					CND Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	301	302	300	299	298	27	27	23	25	25	
	(2)	88	88	87	87	87	46	47	46	46	46	
	(3)	66	66	65	66	66	82	82	83	82	83	
	(4)	58	58	58	58	58	183	185	188	184	187	
(5)	54	54	54	54	54	1184	1550	1770	1637	1055		

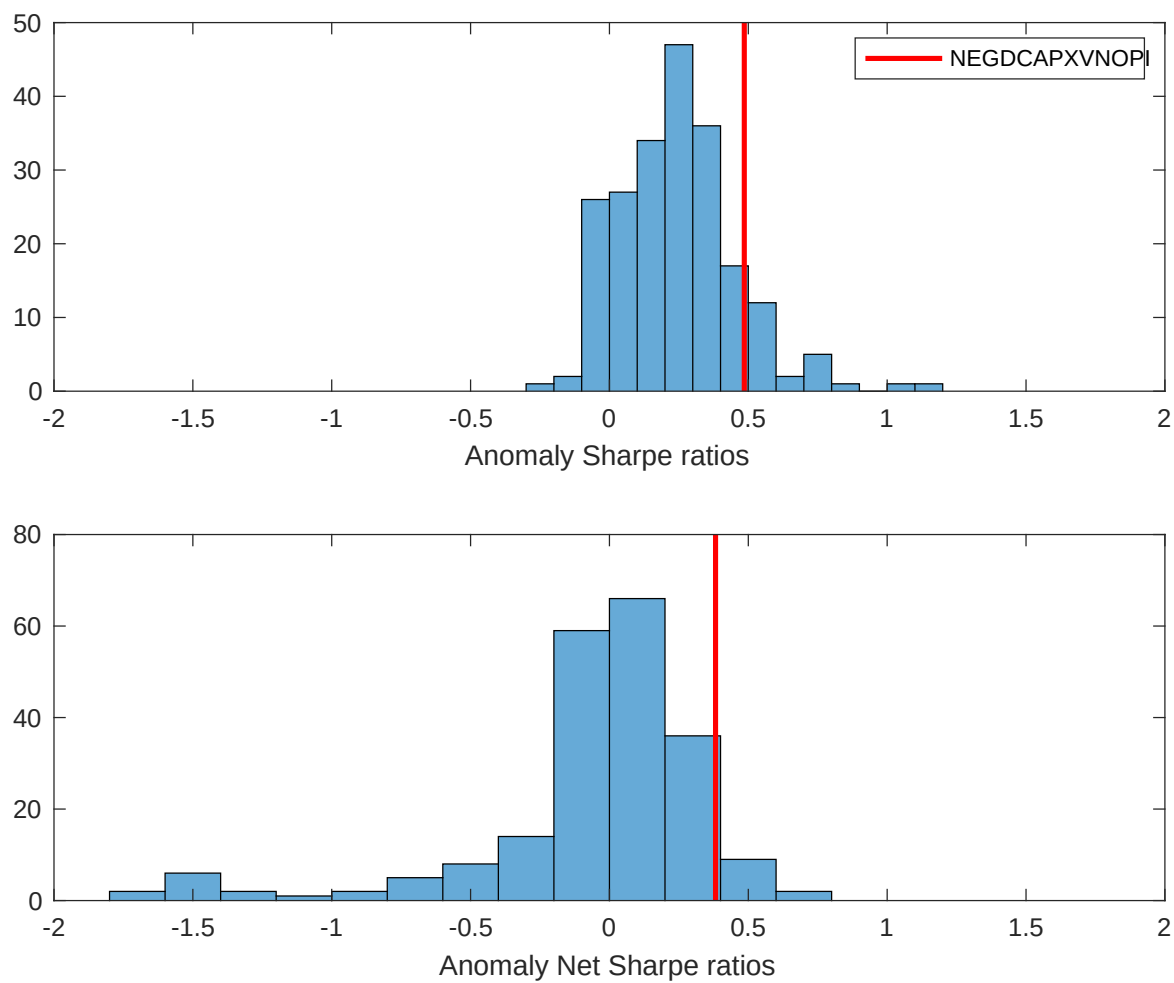


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CND with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

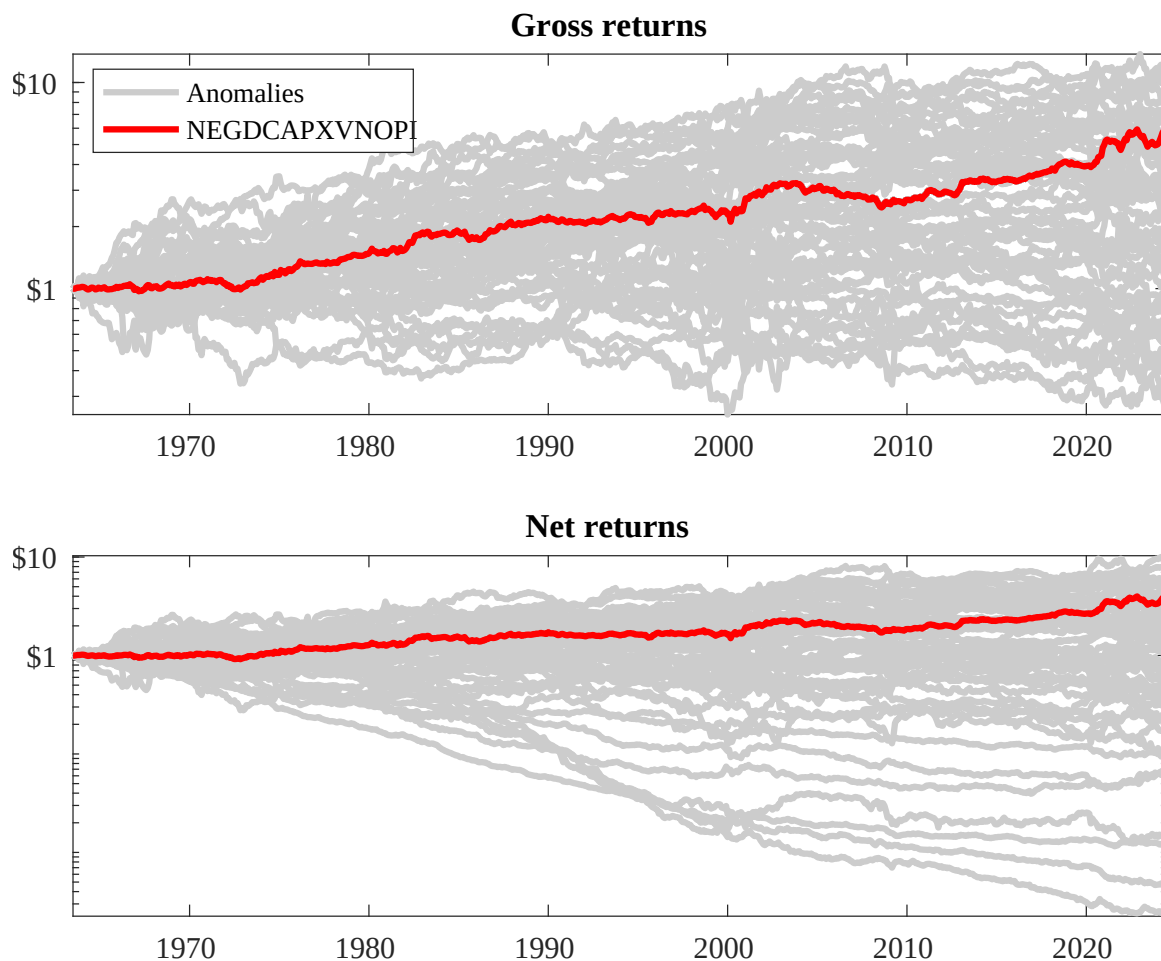
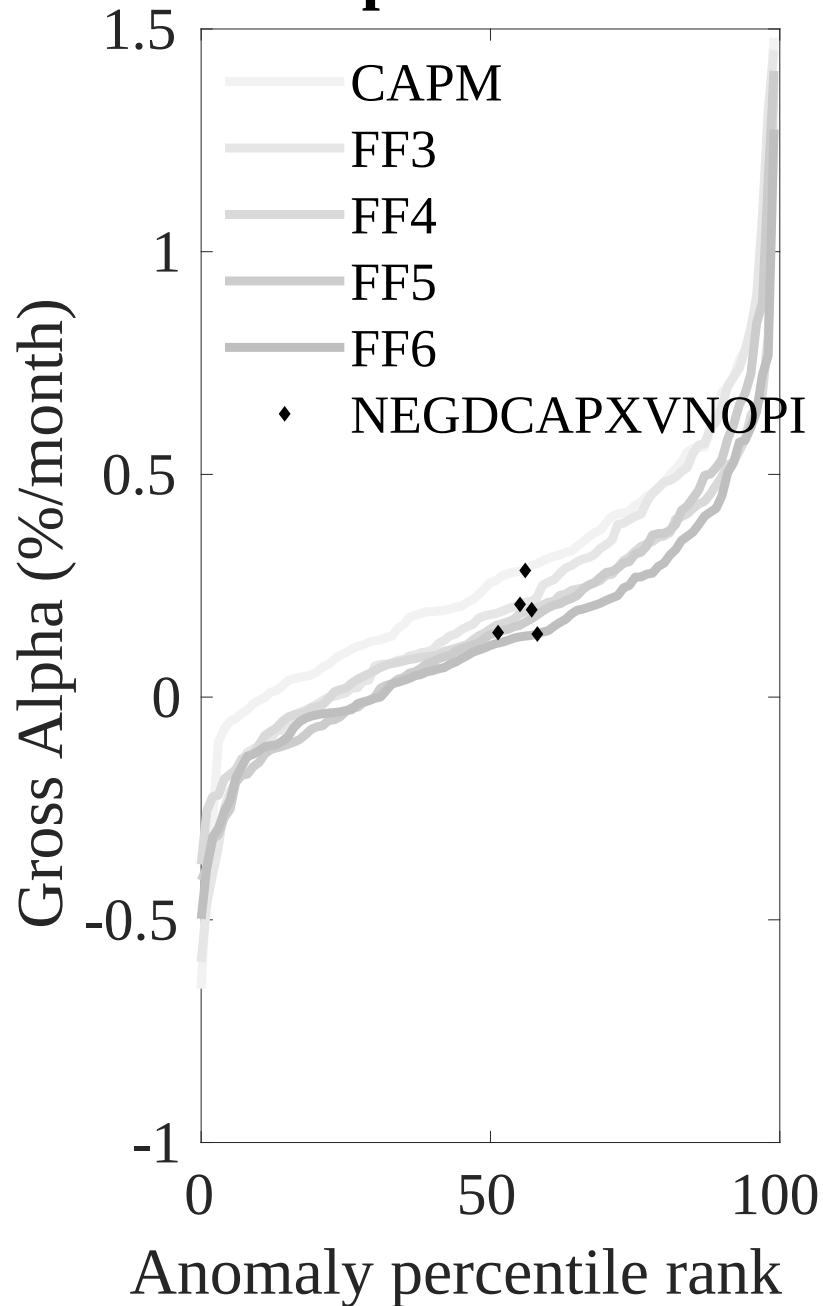


Figure 3: Dollar invested.

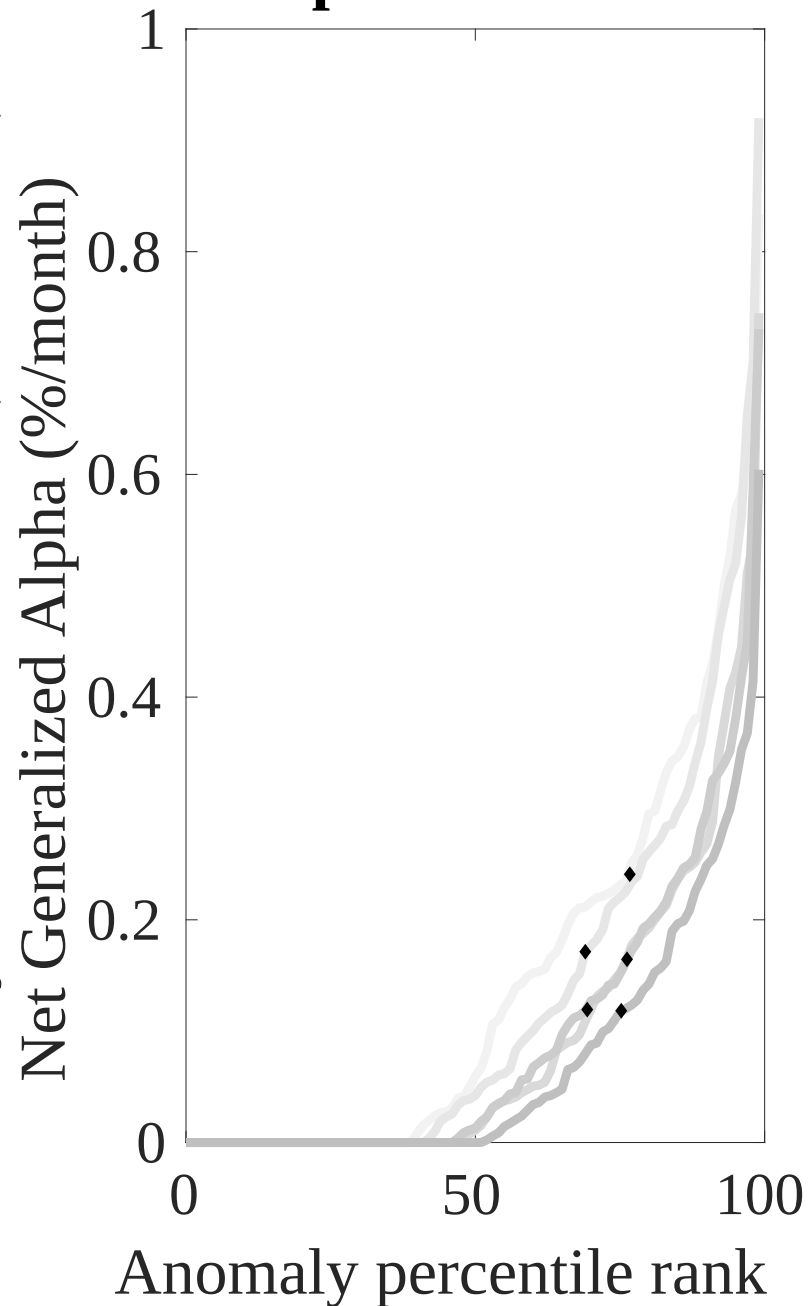
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CND trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



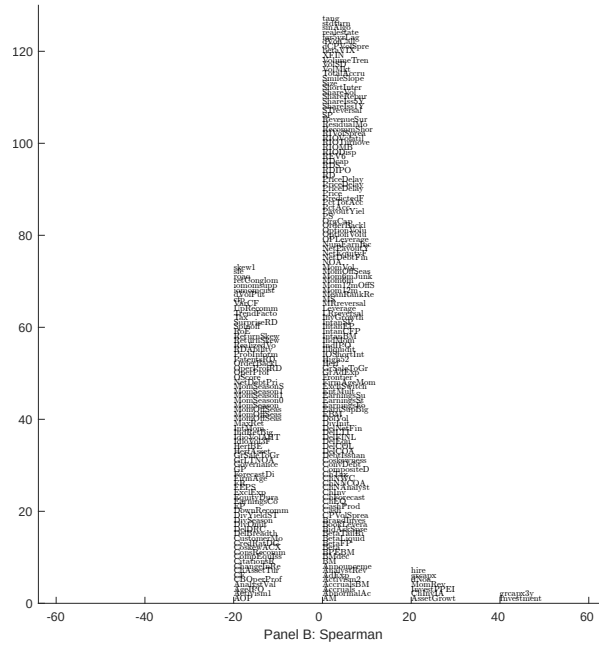
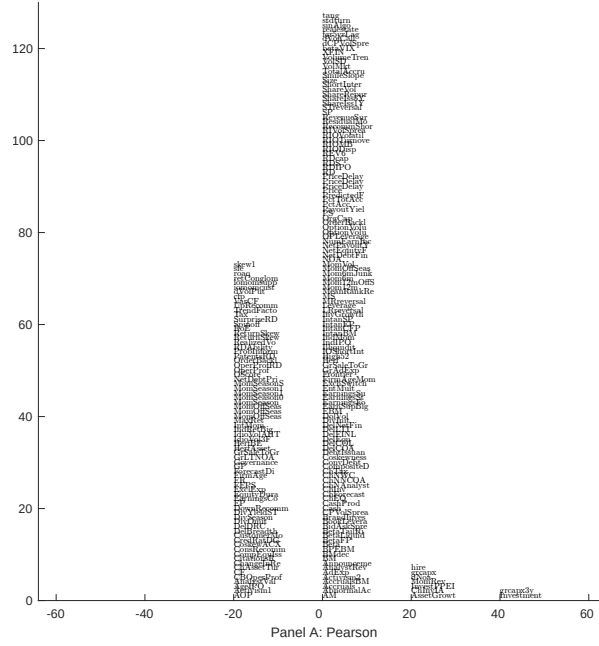


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with CND. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

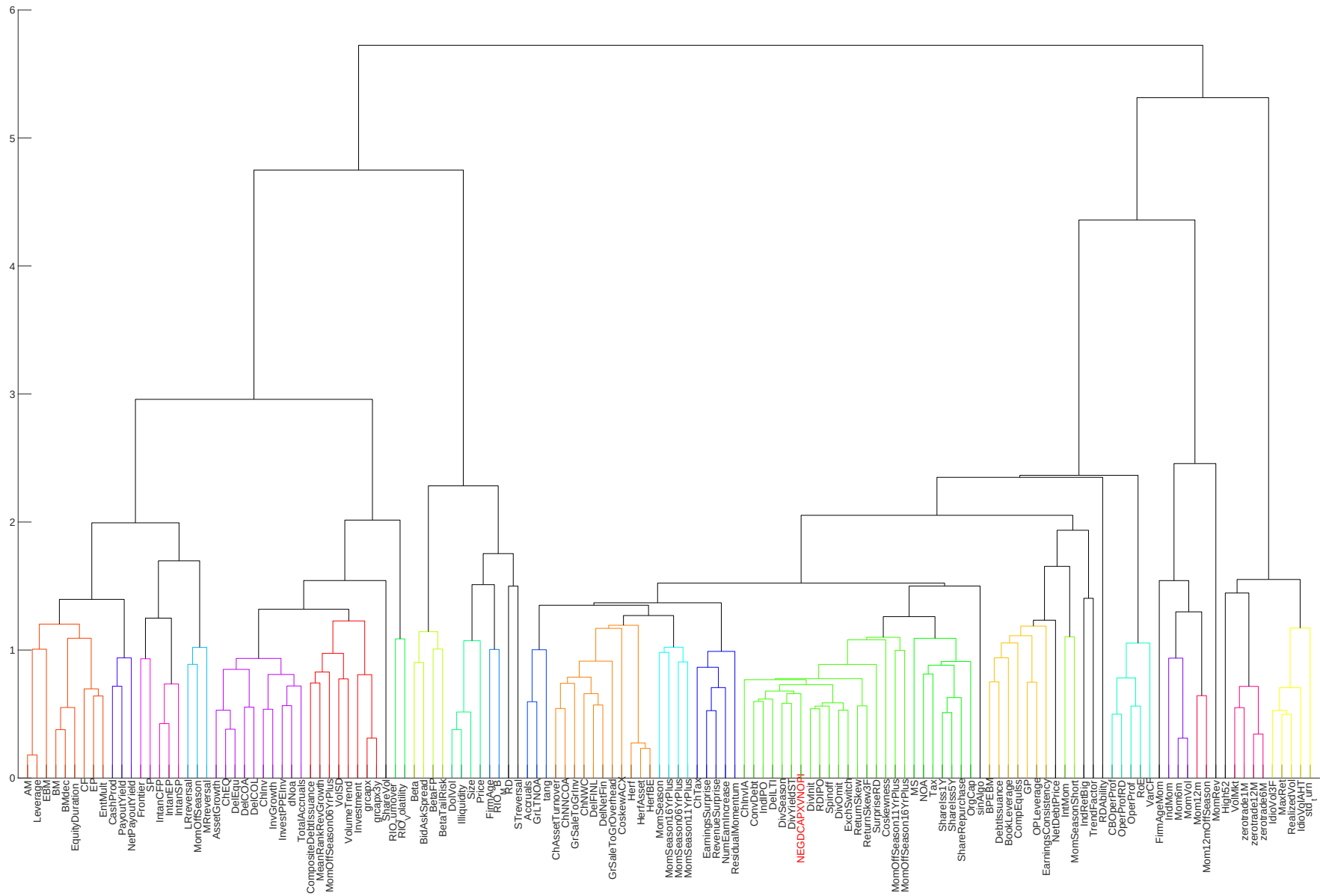


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

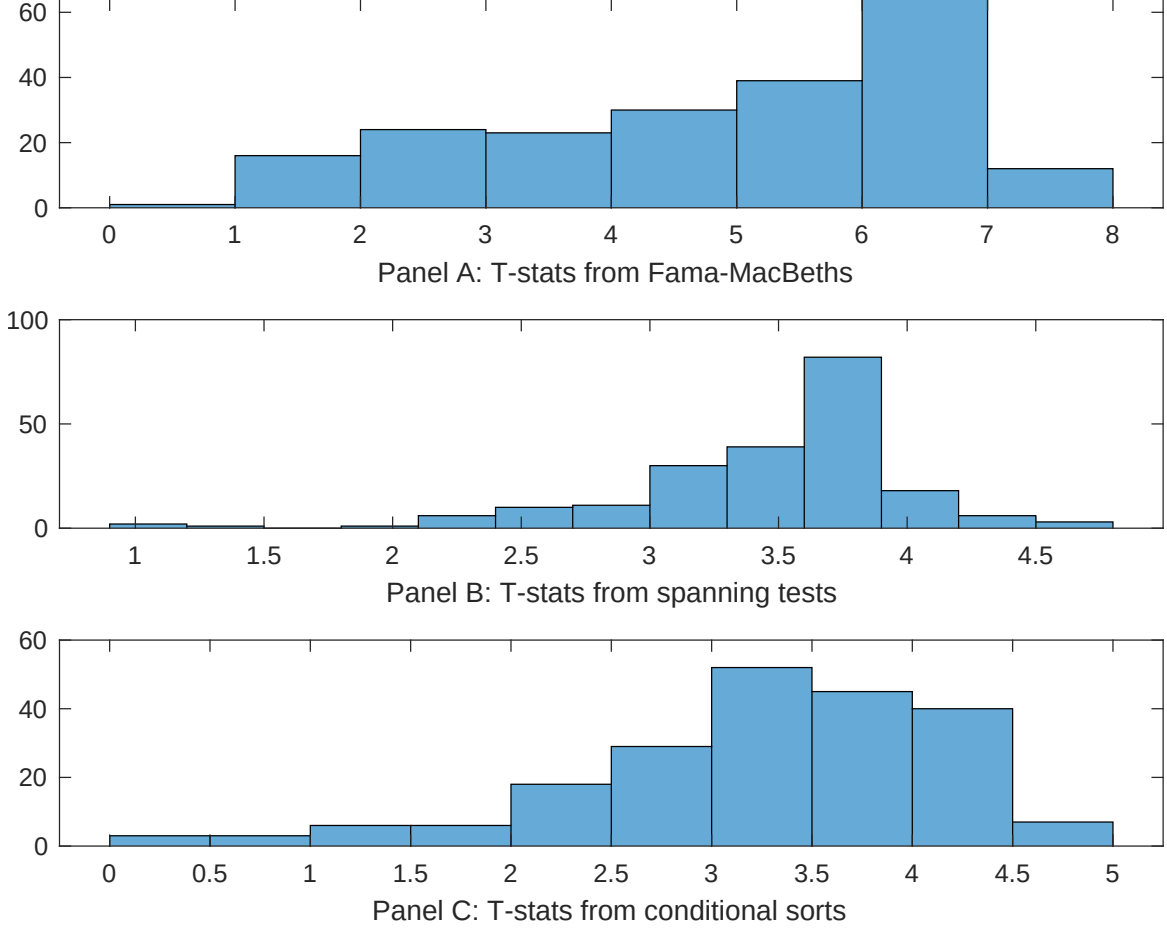


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CND conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CND} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CND}CND_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CND,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CND . Stocks are finally grouped into five CND portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CND trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CND . and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CND}CND_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in capex (three years), Change in capex (two years), Asset growth, change in net operating assets, change in ppe and inv/assets, Change in current operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.14 [6.60]	0.13 [5.86]	0.13 [6.02]	0.13 [5.75]	0.13 [5.99]	0.12 [5.63]	0.14 [6.08]
CND	0.18 [3.42]	0.23 [4.48]	0.16 [3.24]	0.15 [2.97]	0.13 [2.79]	0.29 [5.53]	0.11 [1.96]
Anomaly 1	0.14 [8.00]						0.23 [0.93]
Anomaly 2		0.67 [7.57]					0.18 [1.36]
Anomaly 3			0.96 [8.25]				0.43 [2.21]
Anomaly 4				0.13 [9.09]			0.28 [1.45]
Anomaly 5					0.16 [8.12]		0.80 [0.29]
Anomaly 6						0.21 [6.90]	0.39 [1.06]
# months	727	727	732	720	732	732	715
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CND trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CND} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in capex (three years), Change in capex (two years), Asset growth, change in net operating assets, change in ppe and inv/assets, Change in current operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.15 [2.33]	0.14 [2.23]	0.16 [2.50]	0.12 [1.90]	0.16 [2.40]	0.16 [2.43]	0.14 [2.19]
Anomaly 1	18.75 [5.95]						4.83 [0.86]
Anomaly 2		20.01 [6.35]					10.53 [1.90]
Anomaly 3			26.13 [6.72]				12.88 [3.06]
Anomaly 4				23.05 [6.26]			10.50 [2.59]
Anomaly 5					14.33 [4.62]		6.01 [1.79]
Anomaly 6						13.12 [4.03]	-0.05 [-0.02]
mkt	1.08 [0.71]	1.17 [0.77]	1.71 [1.09]	1.79 [1.14]	1.50 [0.95]	1.59 [1.00]	1.24 [0.82]
smb	-3.45 [-1.50]	-2.21 [-0.99]	-2.09 [-0.92]	0.25 [0.11]	-0.57 [-0.25]	2.74 [1.15]	-3.19 [-1.31]
hml	7.30 [2.41]	7.24 [2.40]	10.61 [3.52]	9.69 [3.19]	9.96 [3.23]	6.59 [1.99]	6.18 [1.96]
rmw	3.66 [1.22]	3.05 [1.02]	5.57 [1.83]	6.13 [2.01]	5.86 [1.90]	7.48 [2.40]	3.66 [1.23]
cma	13.28 [2.85]	12.39 [2.66]	-9.82 [-1.52]	4.53 [0.87]	10.73 [2.11]	14.53 [2.99]	-13.27 [-2.01]
umd	-0.53 [-0.35]	-0.90 [-0.59]	1.33 [0.86]	0.05 [0.03]	0.52 [0.33]	1.09 [0.69]	-0.35 [-0.23]
# months	728	728	732	732	732	732	728
$\bar{R}^2(\%)$	19	20	19	19	17	16	23

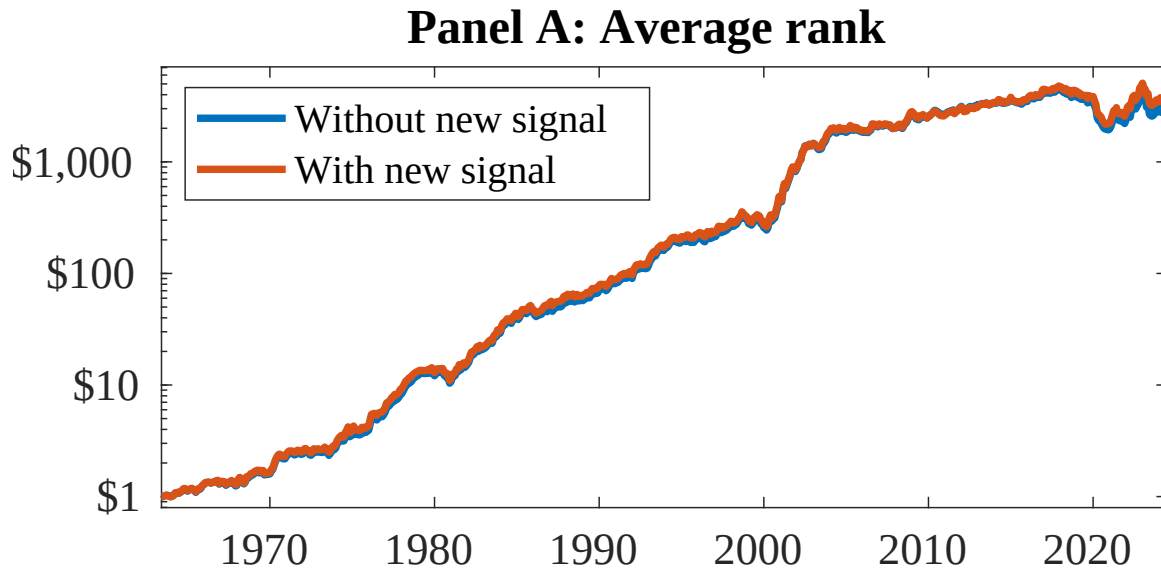


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as CND. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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